Project BluEnergy

Recommendation to the Chairman and the TradingCo investment committee



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Overview of Case Solution

Strategy

OPTION A: dedicated LR1 Jetty (Project Cost is $85M)

Pointers

1. **NPV, IRR, Payback**
2. **Cheaper Cost Procurement**
3. **Diversifying Refinery Sources (more than UGE)**
4. **Tank Farm Restructuring**
5. **New Parent Company (TradingCo) and BluEnergy Arbitrage Opportunities**
6. **Upside Limit Comparisons between Option A and Option B**
7. **Demurrage View**
8. **Risk Picture**
9. **Execution Timeline**

Summary of NPV, IRR, Payback

Question

Where do we deploy $90m? To store more of a product we cannot get cheaply, or to change how we get it?

Strategy

OPTION A : dedicated LR1 jetty · $85m

RECOMMENDED. Removes both structural constraints. Halves freight, ends berth congestion, recommissions the PMEMT line and makes non-UGE origins physically possible.

OPTION B : +250,000 bbl storage · $67m

REJECTED. Fixes ullage and flips the SSPR settlement from paying to being paid, but leaves the berth and the single origin exactly as they are.

Impact

NPV +24.3m · IRR 7.2% · payback 12.7 yrs

NPV -17.7m · IRR 2.9% · payback 16.7 yrs

Assumptions for NPV, IRR, Payback

Key Assumptions	Value
Alternative-origin voyage time	17 days
UGE MR volume share	0% of annual volume
UGE LR1 volume share	70% of annual volume
Alternative-origin MR volume share	0% of annual volume
Alternative-origin LR1 volume share	25% of annual volume
Solar Panel Annual Electricity Saving	USD360,000

+ all the given reference numbers and assumptions given in the case brief (discount rate, GDP growth rate for mining sector etc.)

Every problem in the pack traces to two constraints , the prev. leadership only ever solved around the third

The constraint	Where it hurts	Option A	Option B
BERTH : one shared jetty, first come first-served	$2.1m/yr congestion demurrage. 7% of jetty usage against ~180 container calls.	SOLVED — BluEnergy-exclusive berth, arrivals we schedule ourselves	UNTOUCHED — the $2.1m is permanent
ORIGIN : 100% UGE, MR only, sub-5% lifter	$2.00/bbl freight and a $0.40/bbl small-lifter differential we cannot negotiate away.	SOLVED — LR1 draft opens Pacific and Atlantic; Group book prices the lane	UNTOUCHED — MR economics remain
TANKAGE 500,000 bbl	$1.4m/yr ullage demurrage, all of it in diesel at >80% utilisation.	SOLVED FOR ~$1m — convert one jet tank to diesel	SOLVED FOR $67m — plus $23.8m to fill the tanks

Option B addresses neither constraint that defines this business and buys the third for $67m

We recommend Option A:

Build the dedicated LR1 jetty.

Demurage view.

The $3.5m demurrage bill: $2.1m is solved by owning the berth, $1.4m by not owning the cargo

Demurrage	**CONGESTION : $2.1m (60%)**	**ULLAGE : $1.4m (40%)**
Why it happens	One berth, shared with a container terminal running ~180 calls a year, first come first served. Our tankers arrive to a queue and wait offshore.	We are obliged to take an ENTIRE cargo into a tank farm that cannot always hold it. We buy FOB (free on board) at the refinery, so the barrels are already ours & the ship waits until we have room.
What Option A does	We build our own berth. There is no queue to join. Structural, scheduled timetables to handle this	1 · Buy Ex-ship, so we lift only the parcel we have room for and the ship sails on 2 · Convert one jet tank, 60,000 bbl more diesel headroom 3 · Own the berth, so arrivals are timed into the tank trough (We schedule)

Ullage demurrage exists because we own the cargo. On TradingCo's book, we do not.

TODAY — FOB at the load port

We buy at the refinery in UGE → The barrels are ours for 10 days at sea → We must take the WHOLE cargo → **Tanks full → the ship waits → WE PAY**

PROPOSED — ex-ship at Port Maren

TradingCo buys at the refinery in UGE. It sits on their books, not ours → The LR1 sails full at 450,000 bbl. The barrels are theirs for 10 days at sea → We lift only the parcel our tanks have capacity for → **The ship sails on NO WAITING**

What else it brings

$3.12m a year

We currently lift 7.8m bbl from a refinery that exports 90m Its under 5% of their book, so we pay $0.40/bbl for being small.

$20m off our balance sheet

We stop financing 10 days of cargo at sea. Balance-sheet relief !

Ship size stops being capped by tank size

We can lift off a 450,000 bbl vessel without needing 450,000 bbl of ullage, as we are no longer the only customer on the voyage.

We never own barrels we cannot land → **ullage demurrage disappears**

We buy at TradingCo's scale → **the $0.40/bbl small-lifter premium disappears**

And if the ship has no onward call, the balance is lightered ship to ship at anchorage and moves on from there so the surplus is never stranded on a vessel we are paying for.

Arbitrage Opportunities

Turn Port Maren into an edge-information arbitrage node

- **Solution Statement:**
  - A **new LR1 jetty** plus **direct TeraFuel / mine channels** lets TradingCo convert local physical signals **into cross-network trades** in freight, **crude quality**, **blended products** and **metal ores**.

- **Gaining Access to these specific factors:**

1. Freight Windows
• LR1 vs MR economics • Berth / Ullage constraints
2. Quality Dislocation
• Sweet / Sour premiums • Sulfur specs • Off-spec barrels
3. Demand Pull
• Mine haulage schedules • Fuel tenders • Stock-outs
4. Ore Flow Signals
• Concentrate availability • Port queues • Backhaul demand

450k	6.3m	3.0%
Bbl LR1 parcel	Bbl/yr diesel book	Mining growth (Diesel demand)



This asset is worth more under TradingCo than it could ever be worth standalone - which is the AMBITION the Chairman said was missing

Four oil-only trades BluEnergy unlocks

The same local channel can trigger several TradingCo strategies across ports, terminals and refineries.

1 | Product location arbitrage

Move diesel / Jet A-1 from BluEnergy when Port Maren clears lower than another TradingCo short terminal. Sell the netback, not the local price.

2 | Freight arbitrage

Use LR1 calls, backhauls and berth control to reduce $/bbl freight versus fragmented MR movements through congested shared infrastructure.

3 | Sweet / sour crude routing

Use BluEnergy's demand visibility to redirect sweet or sour crude to the refinery that values its yield slate most, instead of forcing local clearing.

4 | Blend optimization

Blend diesel / gasoil / jet streams to the cheapest compliant spec, then route the barrel to the highest netback terminal in the network.

BluEnergy's role

1. Physical hub: LR1 marine access + storage/testing
2. Information hub: mine-linked fuel demand signals
3. Decision hub: route barrels to highest netback

Trade filter

Execute only when: destination netback − BluEnergy opportunity cost − incremental freight/handling − quality/financing > risk hurdle.

This reframes mine-distributor contacts as early demand and logistics intelligence.

Quant Example: re-route BluEnergy diesel to a higher-netback TradingCo's existing terminal

Field Signal from TeraFuel Channel

- Mine-linked distributor conversations show Port Maren is price-capped near the local benchmark,
- while another TradingCo terminal is short low-sulfur diesel in the same 17-day window.

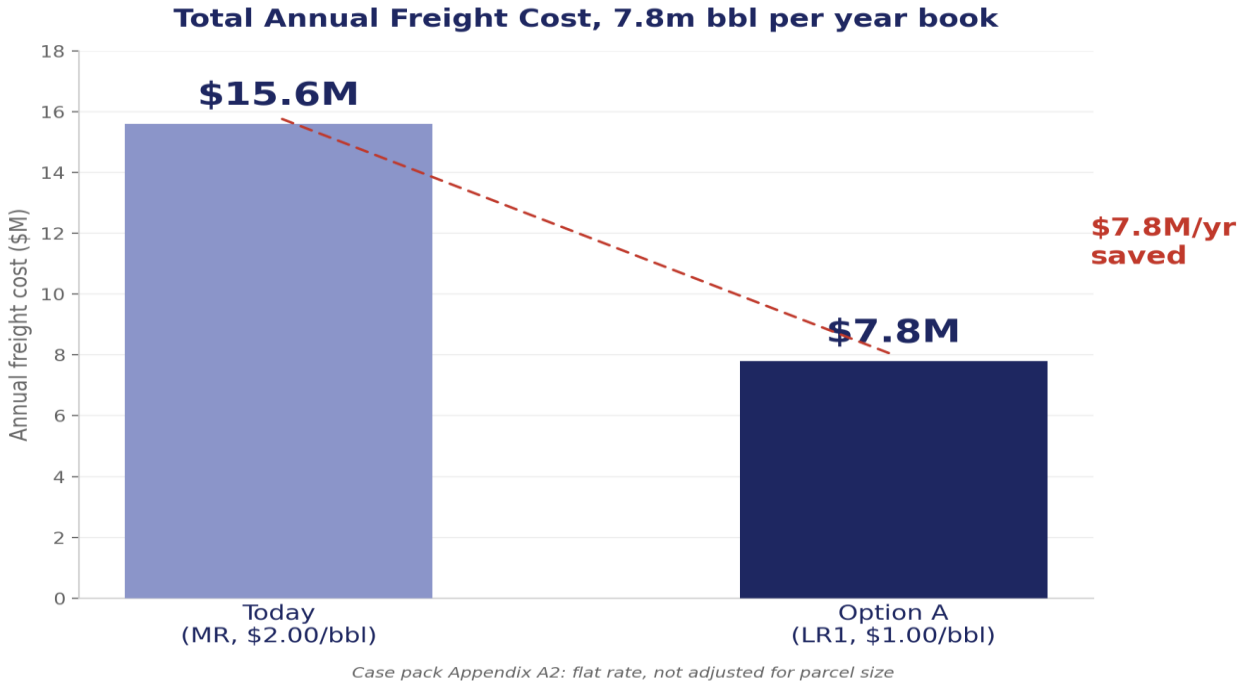


Per Barrel Economics	
Destination sale / netback	$99.10
LESS: BluEnergy Opportunity Cost	($95.00)
LESS: Onward freight + handling	($1.85)
LESS: Quality / Finance / Ops buffer	($0.40)
Net Trading Spread	$1.85 / bbl

Overall Calculations	
Gross Profit per trade (parcel size x net spread)	300k x $1.85 = $0.56M
10 similar trades per annum	$5.6M

Cheaper Cost Procurement

LR1 access cuts freight from $2.00/bbl to $1.00/bbl



Case pack Appendix A2: flat rate, not adjusted for parcel size

LR1 Utilization	Annual Freight Savings ($M)
15% (bear)	1.17
30% (base)	2.34
50%	3.90
75% (bull)	5.85
100% (ceiling)	7.80

The loop Option A closes

1. MR $2.00/bbl → LR1 $1.00/bbl (Appendix A2)
2. On 7.8m bbl/yr: $15.6M today → $7.8M at full LR1
3. At 30% utilization (our assumption): $2.34M/yr saved
4. Flat rate, no parcel-size adjustment, not case-stated

Note. 0 competitors with LR1 access

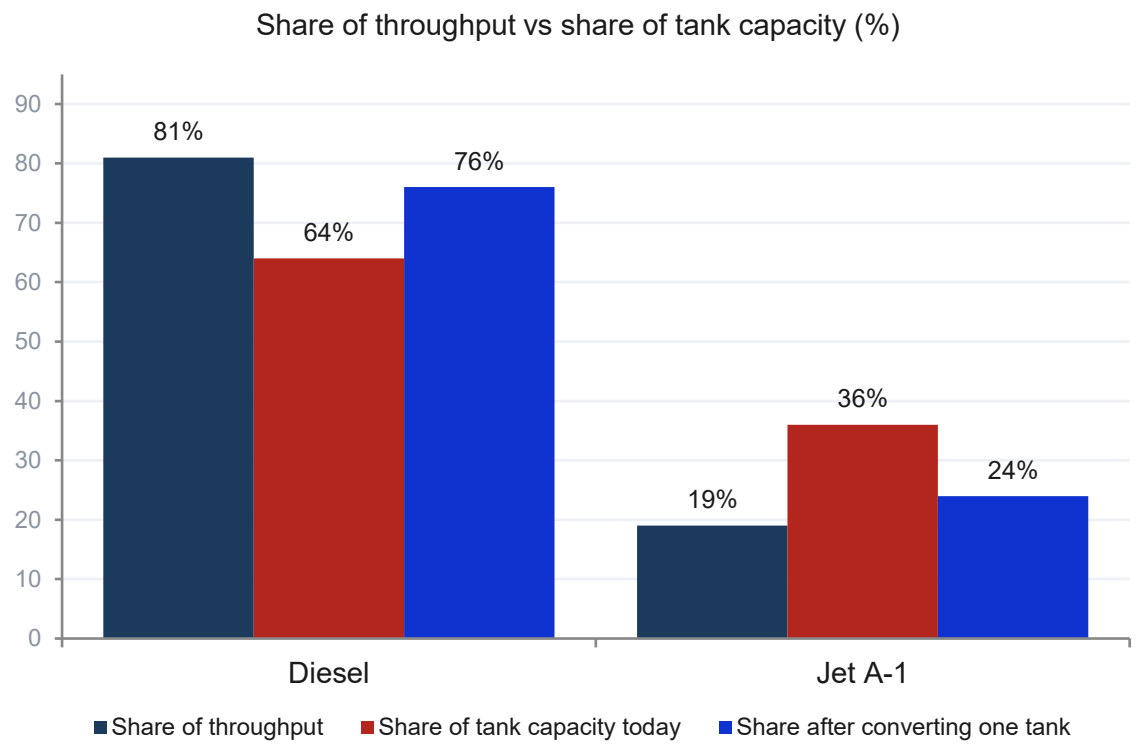
Recommissioning PMEMT Pipeline

- Idle for business reasons, not technical ones
- Recommissioning is already funded, inside Option A's scope
- Faster tank turnover, not more capacity needed
- Fallback if it slips: still 340,000 bbl using tank capacity alone



Tank Farm Restructuring

The tank farm is not too small, it is misallocated. Jet is 19% of volume holding 36% of capacity.



We fix a $1.4m problem for roughly $110k — the previous team proposed to fix it for $67m

Temporary jet-squeeze risk during conversion: jet capacity briefly drops to 120,000 bbl before diesel benefit kicks in **mitigated by scheduling the changeover right after a jet delivery, not before one**

Convert one 60,000 bbl jet tank to diesel	
Capacity	Jet A-1: 180,000 → 120,000 bbl Diesel: 320,000 → 380,000 bbl
Headroom	Diesel: 60,000 → 120,000 Jet: 60,000 → 54,400
Days of cover	Jet: 29 → 16 days (matches diesel's own current 15-day level)
SSPR position	Does not move
Cost	$60K–$160K (industry data), ~14–28 days downtime — fixes a $1.4M/yr problem

Tank reallocation financials

Assumed Demurrage Reduction (of the $1.4M/yr ullage-demurrage baseline)

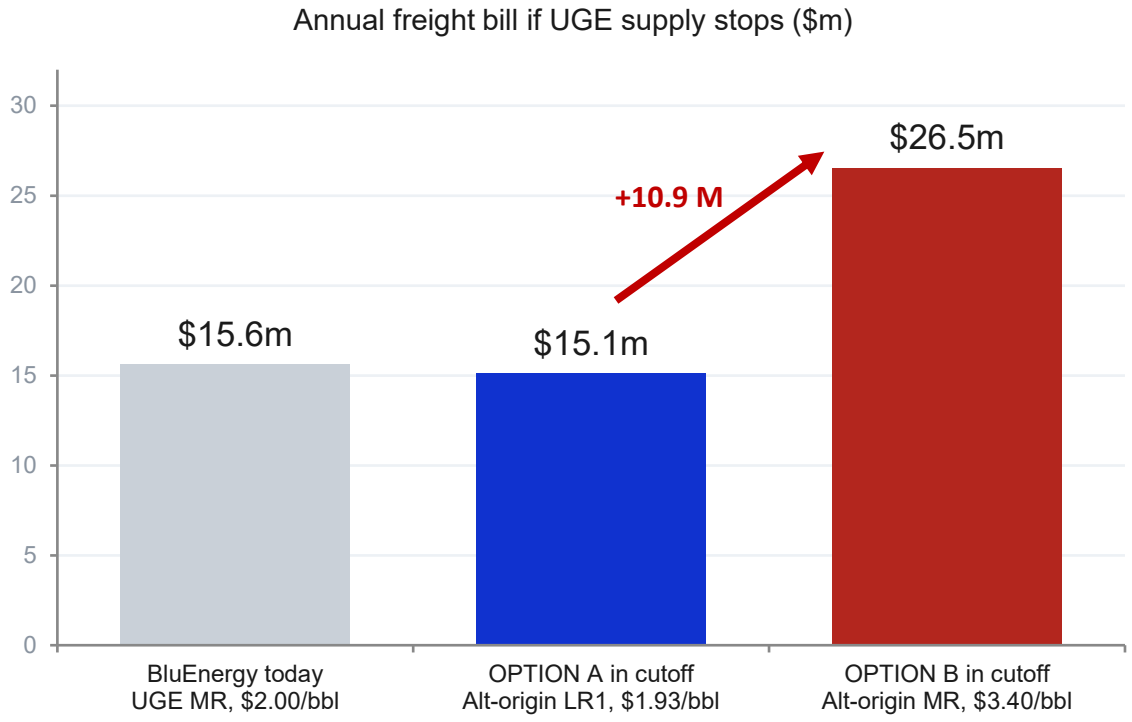
Assumed Demurrage Reductuion	Annual Saving	Payback at $60k (low)	Payback at $110k (base)	Payback at $160k (high)
15%	$210,000	3.4 months	6.3 months	9.1 months
30%	$420,000	1.7 months	3.1 months	4.6 months
50%	$700,000	1.0 months	1.9 months	2.7 months

Every single scenario pays back within a year (add 2 - 4 weeks for downtime)

This fix holds for about a decade — after that, jet's faster growth (4% vs diesel's 1.5%) means we'll need more tank space, and Option A's cash flow pays for it when that time comes.

Diversifying Refinery Sources

In a sudden Gulf cutoff (Geopolitical Tensions), Option A's freight bill is flat against today. Option B's rises 70%.



If UGE supply stops	OPTION A	OPTION B
Replacement vessel	**LR1**	**MR only**
Parcel per call	**~396,000 bbl*** *MWT Current Capacity	**270,000 bbl**
Calls needed per year	**~20**	**~29**
Berth	**dedicated, we schedule**	**Shared, Pay demurrage**
Freight	**$1.93/bbl**	**$3.40/bbl**
Annual freight bill	**$15.1m**	**$26.5m**
vs today's $15.6m	**−$0.5m**	**+$10.9m**

Option A absorbs a total Gulf cutoff for $0.5m less than it pays today. Option B pays $10.9m more.

Freight scaled by voyage days from the pack's $1.00 (LR1) and $2.00 (MR) UGE rates

Option B can charter emergency MR but It arrives slower, in smaller parcels, at double the freight, into the berth that was already generating $2.1m of congestion demurrage in normal conditions.

We are not claiming a cutoff is profitable but we are claiming it is survivable at a freight bill we already absorb today

Capturing TeraFuel's Market-Share

TerraFuels buys from only two suppliers, LR1 lets us win that volume from SNO

Importance to BluEnergy	What is it Worth?
A market with only two possible suppliers. TerraFuels is 28% of the national market, 7.3m bbl/yr, with no import infrastructure of its own, every barrel comes from SNO or BluEnergy.	**Displacement, not subsidy.** Every barrel TerraFuels buys from us is a barrel SNO no longer sells, a direct transfer of existing volume, not new demand we have to create.
No third option exists for TerraFuels. There's no other supplier they can turn to, this is a two horse race for their entire volume.	**A cost edge SNO can't copy without spending.** Matching us costs SNO either margin or a $85m berth it hasn't built, we're competing on structure, not just price.
The exact split isn't given, so we model a range. The case gives no stated SNO/BluEnergy split today, so captured volume should be shown as a range, not a fixed number.	**Margin improves while we undercut SNO's price.** This works because our cost is structurally lower, not because we're accepting a thinner margin to win the volume, we can price below SNO and still come out ahead.

LR1 freight economics are what make undercutting SNO possible without giving up margin. Option B leaves MR economics unchanged, no cost advantage to win this volume with.

Sensitivity Analysis of Terrafuels Market Share Capture

Capture of TerraFuels' 7.3m bbl	Parcel 300,000 bbl	Parcel 340,000 bbl	Parcel 380,000 bbl	Parcel 450,000 bbl
0%	**-6.8**	+16.3	+27.0	+63.6
25%	+26.7	+54.3	+68.6	**+104.6**
50% (base)	+37.9	+70.0	+87.8	**+128.6**
75%	+49.1	+85.7	**+107.0**	**+152.5**
100%	+60.4	**+101.4**	**+126.3**	**+176.4**

Cells are Option A NPV in $m at the pack's 5% discount rate. The base case is 50% capture at a 450,000 bbl parcel. Below 340,000 bbl the tank trough still binds, so only half the ullage fix is credited.

Our assumptions

- Discount rate 5.0%, capex t=0, EBITDA years 2–21 (pack A6)
- Ex-rack margin to TerraFuels $2.00/bbl — ours, not in the pack
- Tank conversion ~$1.0m — ours, not in the pack
- PMEMT transhipment revenue excluded from every case
- New jetty modelled with one discharge arm (§6.1), not two (p.3)

One cell of twenty is negative: we win nothing AND we never get past a 300,000 bbl parcel. The freight rate does not depend on the market share.

Option B is the better project for a standalone BluEnergy. It is not the better project for one owned by TradingCo.

WHAT OPTION B GENUINELY DELIVERS : $5.26m a year

1. **$3.68m a year from the SSPR flip**
   230,000 more barrels held, at $16.00/bbl/yr

2. **$1.4m of ullage demurrage, gone**
   New tanks solve it outright. Under Option we have to schedule for it.

3. **29 days of stock instead of 18**
   All our diesel and jet fuel comes from a single country. Option B buys eleven more days of cover against that..

4. **No execution dependency**
   No PMEMT agreement, no Group supply contract, no berth scheduling required.

WHY WE STILL CHOSE OPTION A

1. **It leaves the berth untouched**
   We keep paying $2.1m a year to wait for a jetty we do not own, with costs growing

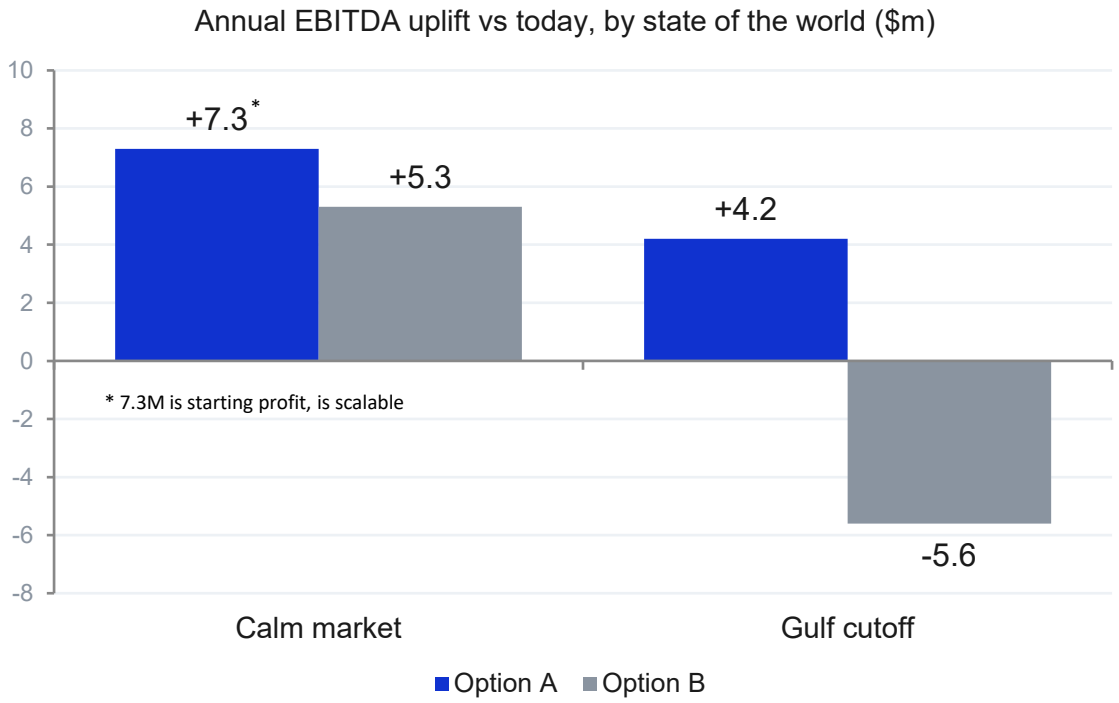
2. **It cannot replace the barrels**
   Storage buys time, not supply. If UGE stops, the alternatives are 19 and 35 days away and those distances only pay on a ship MWT cannot berth. Option B waits 28 days and still has nowhere to buy from. Freight doubles to $26.5m

3. **Option Bs return is capped at about $5.3m**

4. **The real cost of B is $90.8m, not $67m**
   Empty tanks earn nothing. Filling 230,000 barrels of new capacity takes another $23.8m of diesel on day one.

What would change our mind: 3 of Option A's four value drivers (LR1 freight, ex-ship pricing and origin diversification) exist only because TradingCo owns us. If BluEnergy were still an independent company, we would be recommending Option B.

Option A's upside grows with freight, transhipment and market share. Option B's return is fixed

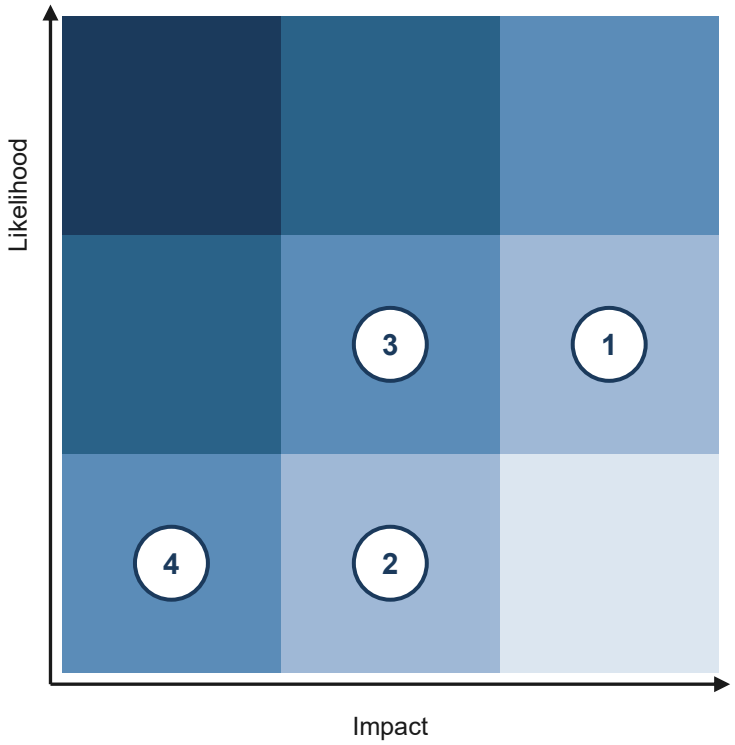


Option B's freight bill doubles in a cutoff (MR only, no LR1 berth) and a fixed $16.00/bbl return cannot absorb it. B swings $10.9m; A swings $3.1m.

	OPTION B	OPTION A
What we receive	$16.00/bbl/yr, fixed	Freight spread, transhipment, market share
Counterparty	The Sarayan government	The market
Ceiling	**Capped at ~$5.3m/yr**	**None**
Floor	**~$5.3m/yr**	**~$7.3m/yr**
In a Gulf cutoff	−$5.6m	+$4.2m
Who else could own it	Anyone with $67m	Only a house that can price three basins

Risk and Mitigation

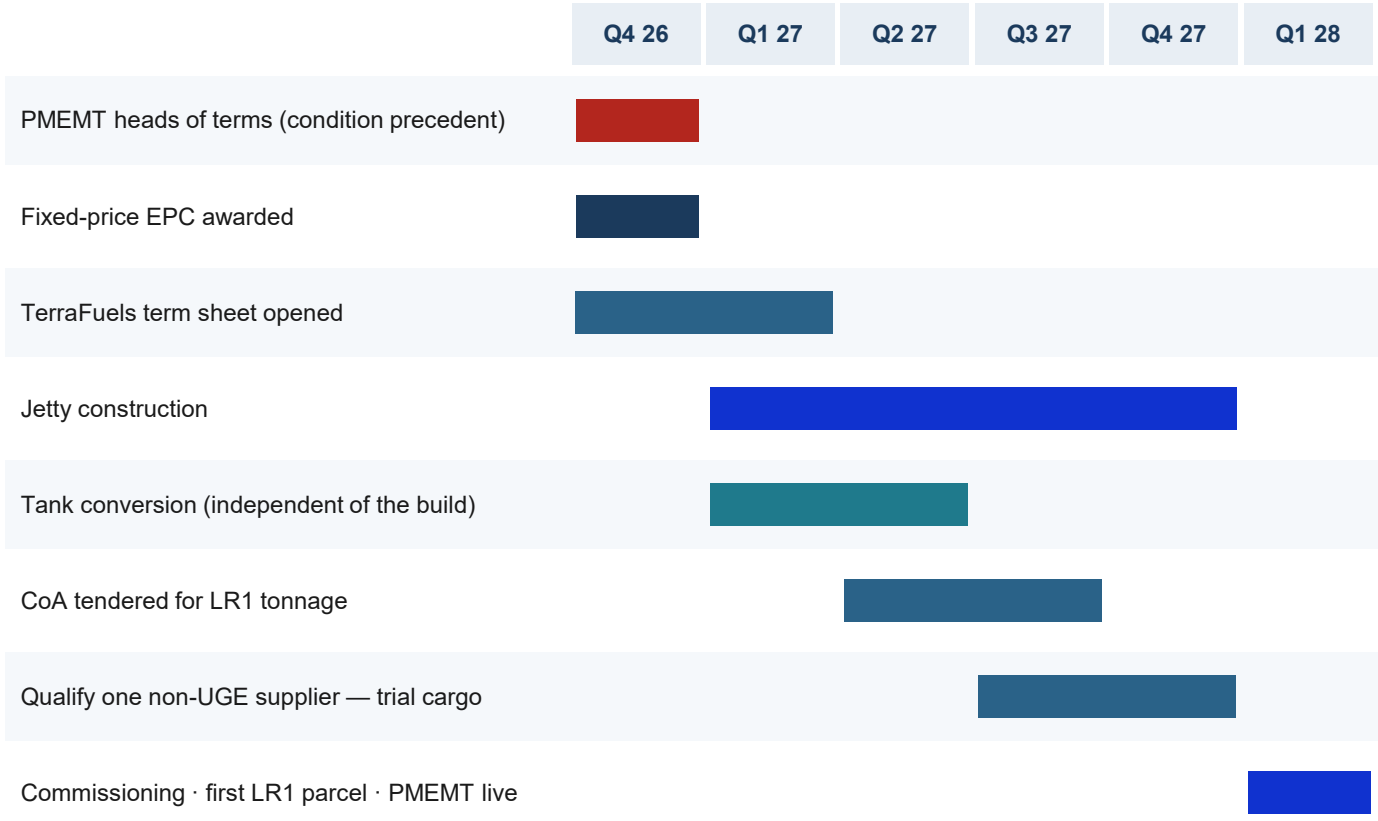
Risk map



	Risk	Mitigation
1	**PMEMT stays a hypothesis** No cross-harbor line, no 450,000 bbl parcel, saving falls to the lower end of the range.	**Mitigation** Signed heads of terms before capital release. Fallback: 340,000 bbl parcels, case still holds.
2	**We win no share from SNO, or the margin is thinner than assumed** TerraFuels volume doesn't arrive, or the $2.00/bbl margin (our estimate) runs thin.	**Mitigation** Already excluded from base case, upside only. CoA guarantees LR1 tonnage regardless.
3	**Jetty overruns on schedule, not just cost** Fixed-price EPC covers cost, not the date. A slip pushes benefits back a year.	**Mitigation** Schedule buffer built into commissioning. Existing jetty stays operational throughout.
4	**Tank conversion squeezes jet, briefly** 14 to 28 day changeover briefly cuts jet capacity before diesel benefit starts.	**Mitigation** Schedule right after a jet delivery, during a low-demand window.

Execution timeline and Implementation considerations

Execution timeline



Implementation considerations

1. Capital release can't hang indefinitely on PMEMT

2. Three commercial negotiations, one small team

3. Fixed-price EPC protects cost, not schedule

4. Road capacity has to be fixed ahead of volume, not after it

5. The tank conversion needs a specific window, not just a slot somewhere in the year

6. The zero-capex wins shouldn't wait for the capex decision to close

None of these six require new capital. What they require is a named owner and a decision date for each, before the clock on the Gantt starts running.

Conclusions

1 **We recommend OPTION A**
The dedicated LR1 jetty — $85m of the $90m authorised.

2 **It is the only option that solves both structural constraints**
One shared berth we do not own; one supply arc we cannot price against.

3 **NPV +$63.6m · IRR 11.2% · payback 8.5 years**
And +$16.3m at IRR 6.8% with zero market-share gain — the growth is upside, not load-bearing.

4 **Demurrage: $2.1m eliminated, $1.3m recovered, $0.1m residual we name rather than hide**
Plus $7.1m/yr of Group plays that need no capital at all.

Option B is an insurance premium. Option A is an option. We do not recommend buying insurance.

Appendix

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A1: How TradingCo operationalizes the edge in 90 days

Make the strategy credible: link field intelligence to trade approval, risk control and measurable P&L.



Idea-generation questions

- Which TradingCo terminal is short this week?
- Which refinery values this sweet/sour barrel most?
- Can BluEnergy blend to a higher-value spec?
- Is LR1, MR split, or backhaul cheaper per bbl?
- Is the spread still positive after quality, demurrage and working capital?

Risk controls for judges

- Execute only on firm quotes or indexed offtake
- Lock freight before lifting cargo
- Pre-clear product spec and tank compatibility
- Cap inventory exposure and hedge flat price
- Track realized P&L by cargo and source signal

Recommendation

Position BluEnergy as TradingCo's Saraya oil hub: a controlled physical node that converts local terminal/distributor intelligence into repeatable oil-product, freight, crude-quality and blending arbitrage.

A2: 8m bbl mining opportunity, next to a metals relationship TradingCo already has

8.0m bbl/yr
Mining diesel demand

2 of 6
Major mines served today

>3% p.a.
Mining growth through 2030

Importance to BluEnergy

1. Mining is Saraya's one durable growth segment left, growing at more than 3% a year through 2030.
2. It is the single largest demand segment in the country,
3. BluEnergy serves only 2 of the 6 major mines today.
4. Mining demand does not carry the substitution risk hitting the rest of the fuel market, unlike the passenger fleet, regulated to zero.
5. Winning share here means growing into a segment, not fighting SNO's 42% share for its own.

What This Is Worth to TradingCo

1. **Berth removes a real constraint on the mines' own exports**
   Metal exports and BluEnergy's fuel cargoes currently share the same jetty. A dedicated berth frees capacity and scheduling reliability for the mines' export shipments.

2. **A real action to point towards, not just a stated relationship**
   This lets TradingCo open with "we already fixed your export bottleneck", a completed action, not a claim still waiting to be tested while expanding to the mines

3. **Positions TradingCo on both sides of the mines' supply chain**
   Supplying diesel in and trading metal out means TradingCo is improving two linked parts of the same relationship, not operating on one side independently of the other.

4. **Priority, not just goodwill**
   Existing nickel and copper relationships with these same six mines. Access, not a guaranteed contract.

Why this only works with Option A

Mines choose suppliers on reliability, and reliability requires the 24/7 marine access, LR1 capability, and self scheduled arrivals only Option A delivers. Option B offers none of it.

A3 · Model conventions, and every number that is ours rather than the pack's

Conventions — all from the pack (A6)

- Discount rate 5.0%
- Full capex at t = 0; no EBITDA during the 12-month build
- 20 years of operational EBITDA, years 2 through 21
- Landed diesel $95.00/bbl, jet A-1 $105.00/bbl, flat
- Diesel demand +1.5% p.a., jet +4.0% p.a. — blended 1.98%
- Real terms: $/bbl savings held flat, volumes grown at the pack's demand rates. Conservative — it ignores 2.5% escalation of the savings themselves.

Assumptions that are ours

Ex-rack margin to TerraFuels	**$2.00/bbl**	Wholesale to a competitor, not the $10.90 blended retail margin
Tank conversion cost	**~$1.0m**	Pack gives none. Jet → diesel is the low-cleaning direction
Jet stock after conversion	**75,000 bbl**	≈18 days on two tanks
Working capital	**40 days of COGS**	FOB pay-at-load + ~10d voyage + receivable
PMEMT throughput fee	**$0.75–1.1m/yr**	No contract exists; revenue from it excluded entirely
Discharge arms on the new jetty	**one**	§6.1 says one, p.3 says two — we took the conservative reading

Nothing on the left is a judgement. Everything on the right is, and each one is sensitised

A4 · Freight is a per-voyage cost — the full derivation behind the $1.00/bbl

The derivation

- Pack A2: MR $2.00/bbl on a ~270,000 bbl parcel → $540,000 per voyage
- Pack A2: LR1 $1.00/bbl on a ~450,000 bbl parcel → $450,000 per voyage
- An LR1 voyage costs ~$450,000 whatever we choose to load
- So effective $/bbl = $450,000 ÷ parcel actually discharged
- The LR1 saves 17% per voyage; the rest of the $1.00 headline is parcel size

Why 450,000 bbl cannot land at MWT alone: diesel capacity after conversion is 380,000 bbl. A 450,000 bbl parcel physically does not fit. The surge crosses the harbour on the 16-inch PMEMT line, which Option A recommissions.

Parcel discharged	Effective $/bbl	Saving vs MR, 7.8m bbl
270,000 bbl	$1.67	$2.60m
300,000 bbl	$1.50	$3.90m
340,000 bbl	$1.32	$5.28m
380,000 bbl	$1.18	$6.36m
420,000 bbl	$1.07	$7.24m
450,000 bbl	**$1.00**	**$7.80m**

340,000 bbl = the Committed case (tankage-constrained after conversion). 450,000 bbl = the base case, requiring the PMEMT line.

Every full-LR1 claim in this deck routes through the PMEMT pipeline — which is why we made it a condition precedent

A5 · The SSPR mechanic, and why growth is not free

Reproducing the pack's own worked example (A4)

Total demand	21,400 bbl/day
25-day requirement	535,000 bbl
Average stock carried	385,000 bbl (~18 days)
Shortfall	150,000 bbl
Settlement at $16.00/bbl/yr	**$2,400,000 paid to the government**

Our model reproduces this exactly, which is our check that the mechanic is right.

Two consequences most models miss

1 · The obligation scales with every barrel we sell

25 days of cover at $16.00/bbl/yr = $1.10 on every barrel of ANNUAL volume. Winning TerraFuels' book raises the requirement from 535,000 to ~784,000 bbl. We deduct it.

2 · Incrementally, the settlement does not decay

Both the base case and the option case face the same growing requirement, so it cancels. The incremental benefit is (extra bbl held × $16) and it is flat. Tightening the obligation from 25 to 40 days would change the narrative and not the number.

Paying the shortfall is rational, not negligent

The shortfall costs less per year than the annualised cost of $67m of tanks. The government has priced an option not to build storage below the cost of building it.

Every barrel we sell ex-rack carries $1.10/bbl of SSPR cost — so we price it in rather than give the freight saving away

A6 · Working capital — the second cost of growth, derived from the FY2025 balance sheet

Derivation

- FY2025 average working capital ~$120m on 7.8m bbl — about $15.40 per barrel of annual volume
- Terms are FOB, pay-at-load, 30 days: we take title and pay at the UGE loading terminal
- The ~10-day voyage and the receivable cycle are financed on our balance sheet
- We model the incremental requirement at 40 days of COGS at $95.00/bbl ≈ $10.40 per barrel of annual volume — deliberately below the observed $15.40
- Funded at TradingCo's blended 5%, not the legacy facility's 10%

Applied to the upside case

Incremental volume at 50% capture	**3.65m bbl**
Working capital required	**$38.0m**
Annual carry at 5%	**$1.90m**
Recovered at the end of the horizon	**yes — modelled at t = 21**

Option B carries the same problem, and cannot avoid it

230,000 bbl of new tankage is worthless empty. Filling it costs $23.8m of product on day one — which is why B deploys $90.8m, not $67m.

Growth consumes capital. We would rather deduct it ourselves than have it found for us

A7 · Option B in full — the version we would have pitched

Option B annual EBITDA build

Ullage demurrage eliminated	**$1.40m**
SSPR settlement flip (pay -> paid)	**$3.68m**
Solar PV	**$0.18m**
Congestion demurrage	—
Freight (MR economics unchanged)	—
Annual EBITDA uplift	**$5.26m**

The sawtooth that drives it

- Average sustainable stock = capacity − cargo ÷ 2
- At 750,000 bbl and a 270,000 bbl MR cargo: 615,000 bbl held on average
- Today: 500,000 − 385,000 held means the trough cannot accept a full parcel — that is literally what the $1.4m of ullage demurrage is
- Incremental stock from B = 230,000 bbl → 230,000 × $16.00 = $3.68m/yr, flat

NPV @ 5%	**-$16.1m**
IRR	**3.2%**
Payback	**17.6 yrs**
Capital deployed	**$90.8m**
Forward cover	**28.7 days**

B's base case does not clear the 5% hurdle the pack sets. We own that rather than torture it

A8 · Our answers, in advance

"Why won't SNO simply cut price?"	They would be matching an $85m structural cost advantage out of margin, indefinitely, as a state-linked entity that has to justify the loss. They cannot match our cost base without their own deep-water berth — 12 months and $85m they have not spent.
"Your share-capture assumption is doing all the work."	It is excluded from the Committed case, which still returns +$16.3m at IRR 6.8%. The freight rate does not depend on the market share.
"PMEMT is a hypothesis, not a contract."	Agreed — your pack says so and we did not argue with it. That is why signed heads of terms are a condition precedent to releasing capital, and why we excluded PMEMT revenue from every case. Without it we take 340,000 bbl parcels and the project still clears.
"You still pay the SSPR settlement."	Yes, and it is rational. The shortfall costs less per year than the annualised cost of $67m of tanks. At $16.00/bbl/yr the government has priced an option not to build storage — below the cost of building it.
"What if the discount rate is higher than 5%?"	Option A's IRR is 11.2% in the base case and 6.8% with no share gain. Option B's is 3.2%. A hurdle above 6.8% would put the Committed case under water, but B fails first and by more.

A9 · Our answers, in advance

"You have assumed a tank conversion cost."	We have, and we flagged it before you asked. At ~$1.0m it is 1.2% of the capex; it would have to be roughly 16× our estimate to change the ranking against Option B.
"What happens to jet cover after you convert a tank?"	18 days on two tanks — ample today. Jet grows at 4.0% a year, so it tightens around 2035. Option A funds new tanks from cash flow well before then. Option B can never fund the jetty.
"What if the Gulf Strait closes?"	Note first that SNO imports through the same arc and TerraFuels has no import infrastructure at all — this is a systemic disruption, not a competitive one. Nobody defects to supply that does not exist. Then: storage lets us wait, the LR1 berth lets us replace.
"Isn't 40% of Option A's value dependent on a berth window you don't control?"	The opposite — today we control nothing, queuing behind ~180 container calls a year. The new berth is BluEnergy-exclusive; the existing shared jetty remains as a back-up. Control of arrival timing is the point of the project.
"Why not split the budget across both?"	The pack does not permit it, and the economics agree: $67m of tanks would consume the budget that makes 450,000 bbl parcels worth landing, and the tank rebalancing delivers most of the ullage benefit for ~$1m.

From the case pack (A6 convention)

Input	Value	Units	Source	Notes
Discount rate	5.0%	%	Case A6	Use unless defended otherwise
Capex timing	t = 0	-	Case A6	Full capex at day one
Operating years modeled	20 (Yrs 2-21)	years	Case A6	No EBITDA during 12-month build
Diesel demand growth	1.5%	% p.a.	Case p.16	
Jet A-1 demand growth	4.0%	% p.a.	Case p.16	
Landed diesel price	$95.00	$/bbl	Case A6	Flat
Landed jet price	$105.00	$/bbl	Case A6	Flat
Saraya inflation	2.5%	% p.a.	Case p.25	

Option A capex inputs (friend's Case Inputs tab, rows 56-66)

Input	Value	Units	Source	Notes
Option A capex	$85,000,000	$	p.19	Jetty + PMEMT refurb + solar canopy
Build duration	12	months	p.19	Commissioning Q1 2027
Nominal LR1 parcel	450,000	bbl/cargo	p.24	
LR1 freight	$1.00	$/bbl	p.24	Future-state if Option A built
Tank capacity after Option A	500,000	bbl	p.19	Unchanged by Option A alone
Dedicated jetty discharge arms	1 arm (case is ambiguous)	arm	p.19	Detail section says 1; exec summary elsewhere implies 2
PMEMT transshipment volume	Not provided	bbl/yr	p.16/19	Commercial arrangement is a hypothesis, not a contract

Option B capacity inputs (friend's Case Inputs tab, rows 33-44)

Input	Value	Units	Source	Notes
Option B capex	$67,000,000	$	p.20	4 tanks + manifold/pipework/firewater + 0.5MW solar
New total capacity	750,000	bbl	p.20	320k+210k diesel; 180k+40k jet
Target forward cover after B	28	days	p.20	
Solar electricity saving	$0.2M/yr	$mm	p.20	0.5MW, ~1.0GWh/yr

Our own flagged assumptions (Assumptions tab, ours)

Input	Value	Units	Source	Notes
LR1 utilization lever	30% (base case)	%	OURS	Tank capacity mismatch means full parcel can't land day one
Demurrage split	60% jetty / 40% ullage	%	Case p.14 (Sec 3.6)	
SSPR requirement	25 days	days	Case	Fixed at $16.00/bbl/yr
Tank conversion cost	$60K-$160K, mid $110K	$	OURS - industry data	Not in the case pack

A2. Freight Rate Comparison

Slide heading: "LR1 Access Cuts Freight from $2.00/bbl to $1.00/bbl"

Scenario	Rate ($/bbl)	Annual Cost / Saving ($M)	Note
Today (MR)	$2.00	$15.6M cost	7.8m bbl x $2.00, Case Appendix A2
Option A (LR1, full adoption)	$1.00	$7.8M cost ($7.8M saved)	7.8m bbl x $1.00, flat rate, not parcel-adjusted

Saving by LR1 utilization level (our Sensitivity tab, rows 5-11)

LR1 Utilization	Annual Freight Benefit ($)	Resulting Option A NPV ($)	
0%	$0	$93,118,061	
15% (bear)	1,171,650	$77,399,965	
30% (base)	2,343,300	$61,681,869	
50%	3,905,500	$40,724,408	
75% (bull)	5,858,250	$14,527,581	
100% (ceiling)	$7,811,000	$11,669,245	

Before / after (our Tank Reallocation tab, rows 21-25)

Metric	Diesel	Jet A-1	Note
Capacity (bbl)	380,000 (was 320,000)	120,000 (was 180,000)	Converts one 60,000 bbl jet tank
Inventory held (bbl)	260,000 (unchanged)	65,600	
Utilization	68.4% (was 81%)	54.7% (was 67%)	
Ullage headroom (bbl)	120,000 (was 60,000)	54,400	
Days of cover	15 (unchanged)	16 (was 29)	Matches diesel's own accepted level

Cost (real industry data, rows 15-17)

Item	Value		
Base cost (midpoint)	$110,000		
Low end	$60,000		
High end	$160,000		
Changeover downtime	14-28 days		Not a case figure

Sensitivity (rows 29-32)

Demurrage Reduction	Annual Saving	Payback (base cost)	Payback range (low-high cost)
25%	$210,000	0.52 yrs	0.3-0.8 yrs
50%	$420,000	0.26 yrs	0.1-0.4 yrs
75%	$700,000	0.16 yrs	0.1-0.2 yrs

Case worked example, reproduced exactly (Appendix A4)

Metric	Value		
Total demand	21,400 bbl/day		
25-day requirement	535,000 bbl		
Average stock carried	385,000 bbl (~18 days)		
Shortfall	150,000 bbl		
Settlement at $16.00/bbl/yr	$2,400,000 paid to government		

Trajectory over 20 years (our Sensitivity tab, rows 16-21)

Operating Year	Option A: SSPR net bbl (worsens)	Option B: SSPR net bbl (erodes)	
1	-160,500	+53,500	
5	-205,483	+8,517	
10	-266,962	-52,962 (flips to shortfall)	
15	-334,840	-120,840	
20	-409,782	-195,782	

Independent cross-check (friend's Option B file, 20Y PnL & DCF, rows 19/110/111)

Metric	Value		
Forward cover at commissioning (Yr 2)	28.0 days		
Forward cover at Year 20	22.29 days		
Option B SSPR settlement, Year 20	+$1.46M/yr (payment)		Confirms the erosion/flip independently

A5. Option A - 20-Year P&L and DCF

Source: friend's file, sheet "20Y PnL & DCF", rows 101-114

Metric	Value	Units	Note
Project capex	85	$mm	
20-year NPV	44.34	$mm	
Project IRR, incl. marine WC	8.87%	%	
Project IRR, excl. marine WC	-6.86%	%	
Simple payback, from t=0	11.34	years	
Simple payback, after commissioning	10.34	years	
Discounted payback @ 5%, from t=0	5.03	years	
Discounted payback, after commissioning	4.03	years	
First operating-year EBITDA uplift	.30	$mm	
Year-20 EBITDA uplift	25.49	$mm	

A6. Option A Sensitivity

Slide heading: "Option A Only Fails in the Case Where We Build the Berth and Never Use It Properly"

LR1 Utilization	Annual Freight Benefit	Resulting Option A NPV
0%	$0	-$93,118,061
15% (bear)	$1,171,650	-$77,399,965
30% (base)	$2,343,300	-$61,681,869
50%	$3,905,500	-$40,724,408
75% (bull)	$5,858,250	-$14,527,581
100% (ceiling)	$7,811,000	$11,669,245

A7. Option B - 20-Year P&L and DCF

Source: friend's file, sheet "20Y PnL & DCF", rows 95-111

Metric	Value	Units	Note
Project capex	67	$mm	
20-year NPV	-17.65	$mm	Negative
Project IRR, incl. inventory WC	2.88%	%	
Project IRR, excl. inventory WC	6.63%	%	
Simple payback, from t=0	16.71	years	
Simple payback, after commissioning	15.71	years	
Discounted payback @ 5%	Not reached	years	
First operating-year EBITDA uplift	.52	$mm	
First operating-year SSPR benefit	.80	$mm	
First operating-year demurrage saving	.53	$mm	
Year-20 EBITDA uplift	.85	$mm	
Year-20 SSPR settlement	.46	$mm	Flipped to a payment
Year-20 stock cover	22.29	days	Down from 28.0 at commissioning

A8. Option A vs Option B, Side by Side

Built from A5 and A7 - not pre-existing in either source file

Metric	Option A	Option B	Note
Capital cost	$85mm	$67mm	
20-year NPV	+$44.34mm	-$17.65mm	Diverges meaningfully
IRR (incl. working capital)	8.87%	2.88%	
Simple payback, from t=0	11.34 years	16.71 years	
First operating-year EBITDA uplift	$6.30mm	$5.52mm	
Year-20 EBITDA uplift	$25.49mm	$9.85mm	

A9. Zero-Capex Group Plays

Our own calculation - not in either Excel model, since these are financing items excluded from both NPVs

Play	Annual Value	Basis
Refinance $80m legacy facility	$4.0M/yr	$80M x (10% - 5%) rate gap
Fold UGE lifting into TradingCo's Gulf book	$3.1M/yr	7.8m bbl x $0.40/bbl small-lifter differential
Combined	$7.1M/yr	More than the entire $3.5M demurrage bill

A18 · Stress-Testing Assumption: Original and Alternative Refinery Station's LR1 and MR allocation – Impact on NPV

Option A — Diversified Supply Scenario Inputs		
Input	**Value**	**Units**
Alternative-origin voyage time	17	days
UGE MR volume share	0%	% of annual volume
UGE LR1 volume share	100%	% of annual volume
Alternative-origin MR volume share	0%	% of annual volume
Alternative-origin LR1 volume share	0%	% of annual volume
Total modeled volume share	100.0%	%

DCF, IRR & Payback Summary			
Metric	**Value**	**Units**	**Interpretation**
Project capex	$85.0	$mm	Spent at t=0
20-year NPV	$44.3	$mm	No terminal value
Project IRR — incl. marine WC	8.9%	%	Unlevered after-tax IRR on incremental FCF, including incremental marine working capital
Project IRR — excl. marine WC	(6.9%)	%	Sensitivity: assumes no incremental marine working-capital burden from the longer voyage
Simple payback — from t=0	11.3	years	Interpolated between annual cash flows; includes 12-month build
Simple payback — after commissioning	10.3	years	Simple payback less the one-year construction period
Discounted payback @ 5% — from t=0	15.0	years	Interpolated using discounted FCF / cumulative NPV
Discounted payback @ 5% — after commissic	14.0	years	Discounted payback less the one-year construction period
PV of operating FCF	$129.3	$mm	Sum of discounted operating cash flows
Undiscounted operating FCF	$243.7	$mm	Before t=0 capex
First operating-year EBITDA uplift	$6.3	$mm	Y2 / Op1
Year-20 EBITDA uplift	$25.5	$mm	Y21 / Op20

IRR / Payback Interpretation	
1	Project IRR ir
2	Simple payba
3	Discounted p
4	The 'IRR excl.
5	Payback after

A18 · IRR / Payback Interpretation

IRR / Payback Interpretation	
1	Project IRR includes the full $85m t=0 capex, zero project EBITDA during the 12-month build, 20 operating years of after-tax incremental cash flow, and no terminal value.
2	Simple payback uses undiscounted incremental free cash flow. It linearly interpolates within the first year in which cumulative project cash flow turns positive.
3	Discounted payback applies the case's 5% discount rate before testing when cumulative NPV turns positive. It is therefore longer than simple payback.
4	The 'IRR excl. marine WC' case is a sensitivity, not the base valuation. It shows how much return is reduced by tying up additional product during the 17-day alternative-origin voyage.
5	Payback after commissioning subtracts the one-year construction period so the team can discuss both 'years from investment' and 'years of operation' in Q&A.